Part III

The fundamental international parity conditions

Part III explains the nature of, and limitations of, two fundamental international financial relationships that between them tie together interest rates, inflation rates, and expected changes in exchange rates. The first of these relationships is the purchasing-power-parity (PPP) principle which occurs in the product markets. It states that, when measured in the same currency, baskets of freely traded goods and services should cost the same everywhere. The second relationship occurs in the money markets, and states that, after allowance for exchange rates, investment yields and borrowing costs should be the same whatever the currency of investment or loan. This relationship is known as the interest-parity principle.

Chapter 7 begins by explaining the law of one price as it applies to an individual commodity such as gold or wheat. In this context the law of one price states that the dollar cost of a particular commodity should be equal everywhere. This means that when prices are measured in local currency, the ratio of, for example, the dollar price of wheat in the United States and the pound price of wheat in Britain should be the exchange rate of dollars for pounds. If not, commodity arbitrage would occur. We show, however, that shipping costs, tariffs, and quotas can result in deviations from the law of one price.

Chapter 7 also explains the extension of the law of one price from an individual commodity to goods and services in general. The extension gives rise to the PPP principle, which states that the ratio of the dollar price of a *basket* of goods and services in the United States to the pound price of the same *basket* in Britain should be the exchange rate of the dollar for the pound.

When considered as a relationship over time rather than at a point in time, PPP becomes a link between changes in exchange rates and differences between inflation rates: currencies of countries with rapid inflation will depreciate vis-à-vis currencies of countries with slow inflation. Chapter 7 explains the nature of this link, and why it might be and, indeed frequently is, broken. It is also shown that another version of PPP that involves expectations of exchange rates and inflation can be derived by considering speculation.

In the context of the money market, which is the market in which short-term securities are traded, the law of one price states that covered (i.e. hedged) dollar rates of return and dollar costs of borrowing will be the same whatever the currency of denomination of the investment or loan. If this were not so, there would be interest arbitrage. This involves borrowing in one currency and lending in another currency, with foreign exchange risk hedged on the forward exchange market. The relationship between interest rates and exchange rates whereby it is irrelevant in which currency a person invests or borrows is called the covered interest-parity condition. This condition is explained in Chapter 8, along with the way that the forward

exchange market can be used to eliminate exchange-rate risk and exposure when engaging in interest arbitrage.

The covered interest-parity condition is derived with the assumptions that there are no transaction costs, political risks of investing abroad, taxes which depend on the currency of investment or borrowing, or concerns over the liquidity of investments. Chapter 8 describes the effects of dropping these assumptions which, as we shall see in later chapters, have bearing on corporate financing decisions and international cash management.

Chapter 7

The purchasing-power-parity principle

> It would be too ridiculous to go about seriously to prove that wealth does not consist in money, or in gold and silver; but in what money purchases, and is valuable only for purchasing.
>
> Adam Smith

Of the many influences on exchange rates mentioned in Chapter 6, one factor is considered to be particularly important for explaining currency movements over the long run. That factor is inflation. In this chapter we examine the theory and the evidence for a long-run connection between inflation and exchange rates. This connection has become known as the **purchasing-power-parity (PPP) principle**. An entire chapter is devoted to the exploration of this principle because it plays an important role in foreign exchange risk and exposure, and many other topics covered in the remainder of this book.

The PPP principle, which was popularized by Gustav Cassell in the 1920s, is most easily explained if we begin by considering the connection between exchange rates and the local-currency price of an individual commodity in different countries.[1] This connection between exchange rates and commodity prices is known as the **law of one price**.

THE LAW OF ONE PRICE

Virtually every opportunity for profit will catch the attention of an attentive individual somewhere in the world. One type of opportunity that will rarely be missed is the chance to buy an item in one place and sell it in another for a profit. For example, if gold or copper was priced at a particular US dollar price in London and the dollar price was simultaneously higher in New York, people would buy the metal in London and ship it to New York for sale. Of course, it takes time to ship physical commodities, and so at any precise moment, dollar prices might differ a little between markets. Transportation costs are also involved in attempts to profit from price differences. However, if there is enough of a price difference between locations, people will take advantage of it by buying commodities in the cheaper market and then sell them in the more expensive market.[2]

People who buy in one market and sell in another are **commodity arbitragers**. Through their

1 Gustav Cassell, *Money and Foreign Exchange after 1914*, Macmillan, London, 1923, and Gustav Cassell, "Abnormal Deviations in International Exchange," *Economic Journal*, December 1918, pp. 413–15.

2 As we shall mention later, one-way arbitrage arguments would mitigate the wedge between prices in different locations caused by transportation costs. If a buyer faced the same transportation costs of sourcing a product from two markets the prices in the two source locations would be driven together.

actions commodity arbitragers remove any profitable opportunities that may exist. They force up prices in low-cost countries and reduce prices where they are high. Normally, arbitragers cease their activities only when all profitable opportunities have been exhausted, which means that except for the costs of moving goods from place to place, including any tariffs that might be involved, prices of the same product in different markets are equal.[3]

In fact, prices of commodities should be the same in different countries even if there is no direct commodity arbitrage between countries themselves. This is because *outside* buyers will select the lowest price. For example, even if it is extremely costly to arbitrage wheat between Canada and Australia, the two countries' prices would need to be the same; otherwise, outside buyers would buy everything from the cheaper and none from the more expensive supplier. In other words, shipping costs between Canada and Australia set only a maximum on the possible price difference between the countries; the actual price difference is generally smaller than this maximum. For example, if Canadian and Australian ports used to export wheat are at the same distance from Chinese receiving ports so that shipping costs to China from Australia and Canada are the same, shipping costs *between* Canada and Australia will not result in different selling prices. Similarly, even when there are tariffs, if they apply equally to potential sources, they cannot cause price differences. In the terminology of Chapter 2, one-way arbitrage creates a tighter link between prices in different countries than does two-way arbitrage.

When prices in different countries are expressed in the *same* currency, the outcome of commodity market arbitrage – that particular commodity prices are everywhere equal – is easily seen. For example, the dollar prices of an ounce of gold in London, Paris, Frankfurt, Zurich, and New York, certainly if observed at the same time with all markets open, are very similar. But what does it mean for arbitrage to ensure that prices are the same when they are in *different* foreign currencies? The answer follows from the **law of one price**, which states that in the absence of frictions such as shipping costs and tariffs, the price of a product when converted into a common currency such as the US dollar, using the spot exchange rate, is the same in every country.[4] For example, because the dollar equivalent of the price of wheat in Britain is $S(\$/£)P^{w}_{UK}$, where P^{w}_{UK} is the pound price of wheat in Britain, the law of one price states that

$$P^{w}_{US} = S(\$/£)P^{w}_{UK} \qquad (7.1)$$

When the law of one price does not hold, buying decisions help restore the equality.

For example, if $P^{w}_{US} = \$4$/bushel, $P^{w}_{UK} = £2.5$/bushel, and $S(\$/£) = 1.70$, then the dollar price of wheat in Britain is $\$1.70/£ \times £2.5/\text{bushel} = \$4.25/\text{bushel}$. With the US price of \$4/bushel, wheat buyers will buy from the US and not from Britain, forcing up the US price and forcing down the British price until they satisfy equation (7.1).

ABSOLUTE (OR STATIC) FORM OF THE PPP CONDITION

If equation (7.1) were to hold for each and every good and service, and we computed the cost of the same basket of goods and services in Britain and the United States, we would expect to find that

$$P_{US} = S(\$/£)P_{UK} \qquad (7.2)$$

P_{US} and P_{UK} are respectively the costs of the basket of goods and services in the US measured in dollars and in Britain in pounds. Equation (7.2) is the **absolute (or static) form of the PPP condition**. The condition in this form can be rearranged to give the spot exchange rate in terms of the relative costs of the basket in the two countries, namely,

$$S(\$/£) = \frac{P_{US}}{P_{UK}} \qquad (7.3)$$

3 We exclude local sales taxes from the price.

4 As we have just said, even in the *presence* of shipping costs and tariffs, the law of one price might still hold as a result of one-way arbitrage.

For example, if the basket costs \$1,000 in the United States and £600 in Britain, the exchange rate according to equation (7.3) should be \$1.67/£.

The PPP condition in the absolute form in equation (7.2) or (7.3) offers a very simple explanation for the level of exchange rates. However, it is difficult to test the validity of PPP in the form of equation (7.2) or (7.3), because different baskets of goods are used in different countries for computing price indexes. Different baskets are used because tastes and needs differ between countries, affecting what people buy. For example, people in cold, northern countries consume more heating oil and less olive oil than people in more temperate countries. This means that even if the law of one price holds for each individual good, price indexes, which depend on the weights attached to each good, will not conform to the law of one price. For example, if heating oil prices increased more than olive oil prices, the country with a bigger weight in its price index for heating oil would have a larger price index increase than the olive-oil-consuming country, even though heating oil and olive oil prices increased the same amount in both countries. Partly for this reason, an alternative form of the PPP condition which is stated in terms of rates of inflation can be very useful. This form is called the **relative (or dynamic) form of PPP**.

THE RELATIVE (OR DYNAMIC) FORM OF PPP

In order to state PPP in its relative (or dynamic) form let us define the following:

> $\dot{S}(\$/£)$ is the percentage change in the spot exchange rate over a year, and $\dot{P}_{US}$ and $\dot{P}_{UK}$ are respectively the percentage annual rates of change in the price levels in the United States and Britain. That is, $\dot{P}_{US}$ and $\dot{P}_{UK}$ are the US and British annual rates of inflation.

If the PPP condition holds in its absolute form at some moment in time, that is,

$$P_{US} = S(\$/£)P_{UK} \qquad (7.2)$$

then at the end of 1 year, for PPP to continue to hold it is necessary that

$$P_{US}(1 + \dot{P}_{US}) = S(\$/£)[1 + \dot{S}(\$/£)] \times P_{UK}(1 + \dot{P}_{UK}) \qquad (7.4)$$

The left-hand side of equation (7.4) is the price level in the United States after 1 year, written as the US price level at the beginning of the year, multiplied by 1 plus the US annual rate of inflation. Similarly, the right-hand side of equation (7.4) shows the spot exchange rate at the end of 1 year as the rate at the beginning of the year multiplied by 1 plus the rate of change in the spot exchange rate. This is multiplied by the price level in Britain after 1 year, written as the price level at the beginning of the year multiplied by 1 plus the British annual rate of inflation. Equation (7.2) is therefore the PPP condition at one point in time and equation (7.4) is the PPP condition a year later.

Taking the ratio of equation (7.4) to equation (7.2) by taking the ratios of the left-hand sides and of the right-hand sides gives by cancelation

$$(1 + \dot{P}_{US}) = [1 + \dot{S}(\$/£)](1 + \dot{P}_{UK}) \qquad (7.5)$$

Equation (7.5) can be rearranged into

$$1 + \dot{S}(\$/£) = \frac{1 + \dot{P}_{US}}{1 + \dot{P}_{UK}}$$

or

$$\dot{S}(\$/£) = \frac{1 + \dot{P}_{US}}{1 + \dot{P}_{UK}} - 1 \qquad (7.6)$$

Alternatively, equation (7.6) can be written as

$$\dot{S}(\$/£) = \frac{\dot{P}_{US} - \dot{P}_{UK}}{1 + \dot{P}_{UK}} \qquad (7.7)$$

Equation (7.7) is the PPP condition in its relative (or dynamic) form.

To take an example, if the United States experiences inflation of 5 percent ($\dot{P}_{US} = 0.05$) and Britain 10 percent ($\dot{P}_{UK} = 0.10$), then the dollar price of pounds should fall, that is, the pound should

depreciate at a rate of 4.5 percent, because

$$\dot{S}(\$/£) = \frac{\dot{P}_{US} - \dot{P}_{UK}}{1 + \dot{P}_{UK}} = \frac{0.05 - 0.10}{1.10} = -0.045, \text{ or } -4.5\%$$

If the reverse conditions hold, with the United States having 10 percent inflation versus 5 percent inflation in Britain, then $\dot{S}(\$/£)$ is positive, and the pound appreciates in value against the dollar by 4.8 percent:

$$\dot{S}(\$/£) = \frac{0.10 - 0.05}{1.05} = 0.048, \text{ or } 4.8\%$$

Both values are close to the 5 percent obtained from taking an approximation of equation (7.7) and writing instead

$$\dot{S}(\$/£) \cong \dot{P}_{US} - \dot{P}_{UK} \tag{7.8}$$

What the PPP condition in this approximate form says is that the rate of change in the exchange rate is equal to the difference between the inflation rates.

Equation (7.8) is a good approximation of equation (7.7) when inflation is low. However, when inflation is high, equation (7.8) may be a poor approximation of equation (7.7). For example, suppose we are interested in the rate of change in the number of US dollars per British pound when British inflation is 25 percent and US inflation is 5 percent. Equation (7.7), the exact PPP formula, implies that

$$\dot{S}(\$/£) = \frac{\dot{P}_{US} - \dot{P}_{UK}}{1 + \dot{P}_{UK}} = \frac{0.05 - 0.25}{1.25} = -0.16, \text{ or } -16\%$$

However, from the approximation based on equation (7.8)

$$\dot{S}(\$/£) = \dot{P}_{US} - \dot{P}_{UK} = 0.05 - 0.25 = -0.20, \text{ or } -20\%$$

Higher inflation makes the approximation even worse. For example, if British inflation was 50 percent; and US inflation 5 percent, then from the precise PPP condition in equation (7.7)

$$\dot{S}(\$/£) = \frac{\dot{P}_{US} - \dot{P}_{UK}}{1 + \dot{P}_{UK}} = \frac{0.05 - 0.50}{1.5} = -0.30, \text{ or } -30\%$$

However, from the approximation

$$\dot{S}(\$/£) = \dot{P}_{US} - \dot{P}_{UK} = 0.05 - 0.50 = -0.45, \text{ or } -45\%$$

The approximation of the dynamic PPP implies a depreciation of the pound much larger than does the exact PPP condition.

The relative or dynamic form of PPP in equation (7.7), and its approximation in equation (7.8), are not necessarily violated by sales taxes or shipping costs that make prices higher than static-form PPP levels. For example, suppose that because of a British value-added tax (VAT) or because of higher shipping costs of principal commodity imports to Britain than to the United States, US prices are consistently lower by the proportion τ than British prices as given by equation (7.2). That is

$$P_{US} = S(\$/£)P_{UK}(1 - \tau) \tag{7.9}$$

Equation (7.9) means that for a given exchange rate and price level in Britain, US prices are only $(1 - \tau)$ of the British level. If the same connection exists 1 year later after inflation has occurred and the exchange rate has changed, we have

$$P_{US}(1 + \dot{P}_{US}) = S(\$/£)[1 + \dot{S}(\$/£)] \times P_{UK}(1 + \dot{P}_{UK})(1 - \tau) \tag{7.10}$$

Taking the ratio of equation (7.10) to equation (7.9) gives

$$(1 + \dot{P}_{US}) = [1 + \dot{S}(\$/£)](1 + \dot{P}_{UK}) \tag{7.11}$$

which is exactly the same as equation (7.5). It follows that equations (7.7) and (7.8), the exact and approximate forms of relative PPP, are unaffected by τ. The relative (or dynamic) form of PPP can hold even if the absolute (or static) form of PPP is (consistently) violated.

EFFICIENT MARKETS (OR SPECULATIVE) FORM OF PPP

The PPP condition, which we have derived from arbitrage considerations, can also be derived by considering market efficiency arguments involving the behavior of speculators.[5] In order to do this let us define the *expected* level and rate of change of the spot exchange rate, and expected inflation in the United States and Britain, as follows:

$S_1^*(\$/£)$ is the expected spot exchange rate in 1 year, and $\dot{S}^*(\$/£)$ is the expected annual percent change in the spot exchange rate. $\dot{P}^*_{US}$ and $\dot{P}^*_{UK}$ are respectively the expected annual rates of inflation in the United States and Britain.

Consider a speculator deciding whether to buy and hold the US basket or the British basket.[6]

After 1 year, each dollar invested in the US basket will be expected to be worth

$$\$(1 + \dot{P}^*_{US}) \tag{7.12}$$

where the dollar sign shows this is a dollar amount. This value will be compared to the expected value of the British basket if it is alternatively bought and held for 1 year. To calculate this, note that each dollar invested in the British basket must first be used to buy pounds, providing[7]

$$£\frac{1}{S(\$/£)}$$

The expected value of this basket in pounds at the end of 1 year is

$$£\frac{1}{S(\$/£)}(1 + \dot{P}^*_{UK})$$

The expected dollar value of this is

$$\$\frac{S_1^*(\$/£)}{S(\$/£)}(1 + \dot{P}^*_{UK}) \tag{7.13}$$

We can write the identity $S_1^*(\$/£) \equiv S(\$/£)[1 + \dot{S}^*(\$/£)]$, and using this in (7.13) allows us to rewrite this as

$$\$[1 + \dot{S}^*(\$/£)](1 + \dot{P}^*_{UK}) \tag{7.14}$$

In an efficient market, and assuming equal risk of speculating in the two baskets, the expected dollar payoffs from holding the British and American baskets must be the same. That is

$$(1 + \dot{P}^*_{US}) = [1 + \dot{S}^*(\$/£)](1 + \dot{P}^*_{UK}) \tag{7.15}$$

Rearranging equation (7.15) to place $\dot{S}^*(\$/£)$ on the left-hand side

$$\dot{S}^*(\$/£) = \frac{(1 + \dot{P}^*_{US})}{(1 + \dot{P}^*_{UK})} - 1 = \frac{\dot{P}^*_{US} - \dot{P}^*_{UK}}{(1 + \dot{P}^*_{UK})} \tag{7.16}$$

Equation (7.16) can be approximated by

$$\dot{S}^*(\$/£) \cong \dot{P}^*_{US} - \dot{P}^*_{UK} \tag{7.17}$$

provided that expected inflation in Britain is not high.

Equation (7.16), and the approximate version in equation (7.17), represent the **efficient markets (or speculative) form of PPP**. While the relative and the efficient markets form of PPP look very similar, they are in fact different. Realized inflation and realized exchange rates may not fit the relative form of PPP at any particular time, but the efficient markets form of PPP should nevertheless still hold. This is because with rational expectations, expected values of variables should on average be equal to realized values. That is, there should be no

5 This approach has been suggested by Richard Roll, "Violations of PPP and their Implications for Efficient Commodity Markets," in Marshall Sarnat and George Szegö (eds), *International Finance and Trade*, Ballinger, Cambridge, MA, 1979.

6 We assume the basket of goods is nonperishable in this version of PPP, or that an index of price levels exists which can be traded.

7 Again, here and in what follows we put the currency symbol in front of magnitudes to show what they are.

persistent biases.[8] This means that if expectations are rational, then on average $\dot{S}^e(\$/£)$, $\dot{P}^*_{US}$ and $\dot{P}^*_{UK}$ should equal the actual rates of change in these variables, so that if equations (7.16) and (7.17) hold, then equations (7.7) and (7.8) also hold, *on average*. Sometimes the change in exchange rates will exceed the values predicted by realized inflation in the countries, and sometimes the change in exchange rate will be less than predicted. On average over a long period of time the over- and under-predictions of exchange rate changes should cancel. We see that speculation, in conjunction with the assumption that expectations are rational, means the relative form of PPP holds on average.

Unfortunately, PPP does not fit the data very well, particularly over short intervals of time. This is because, as we saw in Part II, there are many factors other than commodity prices which influence exchange rates, and these other factors can dominate inflation, at least in the short run. However, rather than simply dismiss the PPP condition as an explanation of exchange rates, let us consider the evidence in more detail to see whether there are circumstances under which PPP gives useful predictions.

THE EMPIRICAL EVIDENCE ON PPP

A major problem in testing the validity of the PPP condition is the need to use accurate price indexes for the inflation rates for the countries being studied. Price indexes cover many items, and what is happening to relative prices within an index is not revealed.

In an effort to use as specific a set of prices as can be obtained and to avoid index-number problems, J. David Richardson considered data on prices of narrowly classified industrial items in the United States and Canada.[9] The classifications are as specific as "cement," "animal feeds," "bakery products," "chewing gum," and "fertilizers." That is, Richardson examined the law of one price rather than the PPP condition. Clearly, if the law of one price does not hold between the United States and Canada, there is not much hope for the PPP condition, especially between countries further apart and more different in the contents of price indexes than are these two neighboring countries.

Richardson estimated an equation similar to that in equation (7.7) for several commodity groups, and found that it did not fit the data well in most commodity categories. Richardson's results suggest that even the law of one price is violated, at least during the time period of his study.

A possible explanation of Richardson's results on the law of one price is the differential pricing of the same object in different countries by multinational firms. Such differential pricing, with higher prices charged where demand is more inelastic, is predicted by the theory of discriminating monopoly.[10] Firms with monopoly power may be able to prevent arbitragers from taking advantage of price differences by withholding supply from any company which handles the monopolists' product and cooperates with arbitragers.[11] This possibility is supported by the observation that where there is little or no opportunity for price discrimination, as in the case of commodity markets, the law of one price does appear to hold in the long run although not in the short run.[12]

Irving B. Kravis and Richard E. Lipsey extensively studied the relationship between inflation rates and exchange rates using different price

8 The lack of bias in expectations if expectations are rational is one of the conditions of rationality described by John F. Muth, "Rational Expectations and the Theory of Price Movements," *Econometrica*, July 1961, pp. 315–35.

9 J. David Richardson, "Some Empirical Evidence on Commodity Arbitrage and the Law of One Price," *Journal of International Economics*, May 1978, pp. 341–51.

10 A monopolist that can segment markets charges a higher price to customers whose demand is relatively price insensitive.

11 Substantial evidence supports the discriminating monopoly argument. See Peter Isard, "How Far Can We Push the 'Law of One Price'?" *American Economic Review*, December 1977, pp. 942–8.

12 See Aris A. Protopapadakis and Hans R. Stoll, "The Law of One Price in International Commodity Markets: A Reformulation and Some Formal Tests," *Journal of International Money and Finance*, September 1986, pp. 335–60.

indexes.[13] They used the GNP implicit deflator (which includes prices of all goods and services in the GNP), the consumer price index, and the producer price index. They also took care to distinguish between goods that enter into international trade (tradable goods) and those that do not (nontradable goods). They discovered, using these many prices and price indexes, that there were departures from PPP. They concluded, "As a matter of general judgment we express our opinion that the results do not support the notion of a tightly integrated international price structure. The record . . . shows that price levels can move apart sharply without rapid correction through arbitrage."[14] They did find that PPP holds more closely for tradable goods than for nontradable goods, but the departures from PPP even over relatively long periods were substantial even for traded goods.

Hans Genberg concentrated on testing to see whether PPP holds more precisely when exchange rates are flexible rather than fixed.[15] The most important aspects of his conclusion can be seen by comparing the two columns in Table 7.1. The table gives the average deviations, in percentages, from an estimated PPP condition. The estimates show departures from PPP for each country with its combined trading partners. The importance of each partner is judged by the share of that partner in the country's export trade. The PPP condition is then statistically fitted between the country, for example, Belgium, and the weighted average of its trading partners. The table shows, for example, that for the United States from 1957 to 1976, the actual difference between the US inflation rate and the inflation rate in its (weighted) trading partners differed from the exchange rate change by, on average, 3.8 percent per annum.

Table 7.1 *Average absolute deviations from PPP, in percent*

	1957–66	1957–76
United States	1.2	3.8
United Kingdom	0.5	3.8
Austria	1.3	2.0
Belgium	1.4	2.1
Denmark	1.3	2.0
France	2.5	3.0
Germany	1.3	2.7
Italy	1.2	5.8
Netherlands	0.5	1.7
Norway	0.9	2.9
Sweden	0.7	1.4
Switzerland	0.7	5.8
Canada	2.0	3.3
Japan	1.9	3.8
Average	1.2	3.2

Notes
There are larger departures from PPP for the years 1957–76, which include a period of flexible exchange rate. However, this could be because conditions were more volatile during the more recent period.

Source: Hans Genberg, "Purchasing Power Parity under Fixed and Flexible Exchange Rates," *Journal of International Economics*, North-Holland Publishing Company, May 1978, p. 260.

The second column of Table 7.1 includes the flexible-exchange-rate period which began in 1973. We find from the average deviations from PPP given at the bottom of the table that the addition of the flexible period makes the deviations increase. The implication is that there were greater violations during the flexible years, 1973–76.[16]

Evidence that considers only years of flexible exchange rates and that is not in agreement with Genberg's conclusion has been provided by Maurice

13 Irving B. Kravis and Richard E. Lipsey, "Price Behavior in the Light of Balance of Payments Theories," *Journal of International Economics*, May 1978, pp. 193–246.

14 Ibid., p. 216.

15 Hans Genberg, "Purchasing Power Parity under Fixed and Flexible Exchange Rates," *Journal of International Economics*, May 1978, pp. 247–67.

16 Genberg also discovered that most of the departures from PPP resulted from movements in exchange rates rather than from changes in price levels. This supports Richardson's conclusion. See also Mario Blejer and Hans Genberg, "Permanent and Transitory Shocks to Exchange Rates: Measurement and Implications for Purchasing Power Parity," unpublished manuscript, International Monetary Fund, 1981.

Obstfeld.[17] Using data for the OECD countries over the 20-year period 1973–93, Obstfeld finds strong evidence for the relative form of PPP. Similar support for the principle in this form, but from a very different data set, is obtained in the study of German hyperinflation by Jacob Frenkel.[18] Indeed, Frenkel finds that in general, every 1 percent difference in inflation is associated with a 1 percent change in the exchange rate, as predicted by PPP. It would be surprising in the extreme circumstances experienced during hyperinflation if we did not find agreement with PPP.

Niels Thygesen has summarized the results of a study by the Commission of the European Communities, which set out to discover how long it takes for inflation rates to restore PPP after exchange rates have been artificially changed by the government to gain competitiveness for exports.[19] The idea is that devaluation should raise the rate of inflation until PPP is restored. This could come about via higher import prices and consequent wage demands setting off reactions elsewhere in the economy. Using economic models of Britain and Italy, the study concluded that it took 5–6 years for inflation differentials to restore the PPP condition. However, Thygesen also observed that 75 percent of the return to PPP was achieved within 2 years.

The question of whether and how long it takes for exchange rates to return to PPP levels after departing from them concerns **reversion to the mean**. Niso Abuaf and Philippe Jorion directly studied this issue of reversion using 80 years of data, a period long enough to offer the prospects of seeing if there is a long-run tendency towards parity.[20] They find that in the event of a 50 percent overvaluation of exchange rates relative to PPP, it takes 3–5 years for the departure to be halved.

Another study that examined how long it takes for PPP to be restored after being disturbed is that of John Hodgson and Patricia Phelps.[21] They used a statistical model that allows lags and discovered that differential inflation rates precede the change in exchange rates with a lag of up to 18 months. A similar conclusion was reached by William Folks, Jr, and Stanley Stansell.[22] Their purpose was to forecast changes in exchange rates, and they discovered that exchange rates do adjust to relative inflation rates, but with a long lag.

A conclusion that differs from that of Hodgson and Phelps and Folks and Stansell was reached by Richard Rogalski and Joseph Vinso.[23] They chose a flexible-exchange-rate period, 1920–24, and studied relative inflation for six countries.[24] Rogalski and Vinso concluded that there is no lag. This, they claim, is what is expected in an efficient market, because relative inflation rates are publicly available information and should therefore be reflected in market prices such as exchange rates. This question of efficiency in the spot exchange rate has been tackled by Jacob Frenkel and Michael Mussa, who argue that even if we do observe departures from PPP, this does not imply that foreign exchange

17 Maurice Obstfeld, "International Currency Experience: New Lessons and Lessons Relearned," *Brookings Papers on Economic Activity*, No. 1, 1995, pp. 119–220.

18 Jacob A. Frenkel, "Exchange Rates, Prices and Money: Lessons from the 1920s," *American Economic Review*, March 1980, pp. 235–42.

19 Niels Thygesen, "Inflation and Exchange Rates: Evidence and Policy Guidelines for the European Community," *Journal of International Economics*, May 1978, pp. 301–17.

20 Niso Abuaf and Philippe Jorion, "Purchasing Power Parity in the Long Run," *Journal of Finance*, March 1990, pp. 157–74.

21 John A. Hodgson and Patricia Phelps, "The Distributed Impact of Price-Level Variation on Floating Exchange Rates," *Review of Economics and Statistics*, February 1975, pp. 58–64.

22 William R. Folks, Jr. and Stanley R. Stansell, "The Use of Discriminant Analysis in Forecasting Exchange Rate Movements," *Journal of International Business Studies*, Spring 1975, pp. 33–50.

23 Richard J. Rogalski and Joseph D. Vinso, "Price Level Variations as Predictors of Flexible Exchange Rates," *Journal of International Business Studies*, Spring 1977, pp. 71–81.

24 This period was also studied by Jacob A. Frenkel, not because exchange rates were flexible but because the inflationary experience was so extreme. See Jacob A. Frenkel, "Purchasing Power Parity: Doctrinal Perspective and Evidence from the 1920s," *Journal of International Economics*, May 1978, pp. 169–91.

markets are inefficient. Exchange rates, they show, move like stock and bond prices. Indeed, Frenkel and Mussa find average monthly variations in exchange rates to be more pronounced than the variation of stock prices.[25]

As we have mentioned, a possible reason for departures from PPP is the use of different weights in different countries' baskets of goods. This possibility is considered in the careful work of Irving Kravis *et al.* (1975).[26] To overcome the problem of different weights they recalculated foreign inflation using US weights for all countries' price indexes, rather than own-country weights. They also separated data according to whether the items were traded or nontraded. Far stronger support for the PPP condition was found in the common-weight inflation data, especially for traded goods. However, since the usefulness of PPP for explaining exchange rates hinges largely on it applying to broad baskets of goods and using available price indexes based on local consumption patterns, we are left to conclude that PPP provides a rather limited description of exchange-rate levels and changes.

Our conclusion to the empirical evidence, that PPP violations do occur, should come as little surprise. Those who travel extensively often observe that PPP does not occur. There are countries that travelers view as expensive and others that are viewed as cheap. For example, Switzerland and Japan are generally viewed as expensive, while India and China are viewed as relatively cheap. This indicates, without any formal empirical evidence, that there are departures from PPP, at least in the absolute or static form. There are two major reasons we can offer as to why this occurs.

25 Jacob Frenkel and Michael Mussa, "Efficiency of Foreign Exchange Markets and Measures of Turbulence," *National Bureau of Economic Research*, Working Paper 476, Cambridge, MA, 1981.

26 Irving Kravis, Zoltan Kenessey, Alan Heston and Robert Summers, *A System of International Comparisons of Gross Product and Purchasing Power*, Johns Hopkins University, Baltimore, 1975.

REASONS FOR DEPARTURES FROM PPP

Restrictions on movement of goods

The possibility of two-way arbitrage allows prices to differ between markets by up to the cost of transportation. For example, if it costs \$0.50/bushel to ship wheat between the United States and Canada, the price difference must exceed \$0.50 in either direction before two-way arbitrage occurs. This means a possible deviation from the absolute form of PPP for wheat that spans \$1. In reality, however, competitive pressures for similar prices to buyers in other countries will keep prices in a narrower range than would result from two-way arbitrage between Canada and the United States. This is the one-way arbitrage referred to earlier in this chapter.

Import tariffs can also cause PPP violations. If one country has, for example, a 15-percent import tariff, prices within the country will have to move more than 15 percent above those in the other before it pays to ship and cover the tariffs that are involved. The effect of tariffs is different from the effect of transportation costs. Tariffs do not have a symmetric effect. As a result of tariffs, price levels can move higher only in the country which has the import tariffs.

Whether it be transportation costs or tariffs, these factors explain departures from PPP only in its absolute or static form. As indicated earlier in the derivation of equation (7.11), when the maximum price difference from shipping costs and import tariffs has been reached, the PPP principle in its relative or dynamic form should explain movements through time that push against the maximum price difference. For example, suppose that prices at existing exchange rates are already 25 percent higher in one country because of import tariffs or transportation costs. If that country has an inflation rate that is 10 percent higher than the inflation rate in another country, its exchange rate will have to fall, on the average, by 10 percent to prevent commodity arbitrage.

Quotas, which are limits on the amounts of different commodities that can be imported, generally

mean that price differences can become quite sizable, because commodity arbitragers are limited in their ability to narrow the gaps. Like import tariffs, they provide a reason for persistent departures from PPP.

Price indexes and nontraded outputs

We have already observed in describing the research of Irving Kravis and Richard E. Lipsey that many of the items that are included in the commonly used price indexes do not enter into international trade. We cannot, therefore, invoke the notion of commodity arbitrage to create an equivalent of equation (7.1) for these items. Most difficult to arbitrage between countries are immovable items such as land and buildings; highly perishable commodities such as fresh milk, vegetables, eggs, and some fruits; and also services such as hotel accommodations and repairs. These "nontraded" items can allow departures from PPP to persist when we measure inflation from conventional market-basket price indexes.

To some extent, a tendency towards parity even in nontraded items can be maintained by the movement of the buyers instead of the movement of the items themselves. For example, factories and office complexes can be located where land and rent are cheap. Vacationers can travel to places where holidays are less expensive. The movement of *buyers* tends to keep prices in different countries in line with each other.

The relative prices of traded versus nontraded outputs will not differ greatly between countries if *producers* within each country can move into the production of the nontraded outputs when their relative prices increase. Consequently, if comparative advantages do not differ significantly between nations, the relative prices of traded versus nontraded items will be kept similar between countries by the prospective movement of domestic producers. Therefore, if the prices of traded goods satisfy PPP, then so should the prices of nontraded items. However, we do require that the producers can move between traded and nontraded goods, and that comparative advantage differences are small. And even when producers can move, price adjustment can take a very long time, during which departures from PPP can persist.[27]

STATISTICAL PROBLEMS OF EVALUATING PPP

It has been suggested that the difficulty finding empirical support for PPP may be due to the statistical procedures that have been used.[28] We can indicate the problems with the statistical procedures by examining the bases for judging empirical support for PPP.

Most tests of PPP are based on estimates of a regression equation that in the context of the dollar–pound exchange rate can be written

$$\dot{S}(\$/£)_t = \alpha + \beta\left[\frac{\dot{P}_{US} - \dot{P}_{UK}}{1 + \dot{P}_{UK}}\right]_t + \varepsilon_t \qquad (7.18)$$

where ε is the *ex ante* regression error and the subscripts t refer to the time period of observation of the variables. It is argued that if PPP is valid, then in estimates of equation (7.18), α should be close to zero, β should be close to 1.0, and the *ex post* regression errors should be small.[29] This is because

27 Mario Blejer and Ayre Hillman have provided a formal model with the costs of commodity arbitrage allowing temporary departures from PPP. See their article "A Proposition on Short-Run Departures from the Law of One Price: Unanticipated Inflation, Relative Price Dispersion, and Commodity Arbitrage," *European Economic Review*, January 1982, pp. 51–60.

28 The statistical problems have been surveyed by John Pippenger, "Arbitrage and Efficient Markets Interpretations of Purchasing Power Parity: Theory and Evidence," *Economic Review*, Federal Reserve Bank of San Francisco, Winter 1986, pp. 31–48. See also Sandra Betton, Maurice D. Levi, and Raman Uppal, "Index-Induced Errors and Purchasing Power Parity: Bounding the Possible Bias," *Journal of International Financial Markets, Institutions and Money*, 2/3, 1995, pp. 165–179.

29 Small errors mean that the equation fits well. The "goodness of fit" measure that is usually used is the R^2 statistic, which gives the fraction of the variation in the dependent variable, $\dot{S}(\$/£)$, that is explained by the explanatory variable(s), in this case, $(\dot{P}_{US} - \dot{P}_{UK})/(1 + \dot{P}_{UK})$.

in such a case the regression equation reduces to equation (7.7). The statistical problems that can result in incorrect rejection of PPP are:

1 *Errors in measuring the inflation differential* A characteristic of the regression methodology is that errors in the measurement of explanatory (i.e. right-hand) variables bias regression coefficients towards zero.[30] This means that if the inflation differential is poorly measured because different baskets are used in each country, then we could find the estimated β to be smaller than 1.0 even if the true β is exactly equal to 1.0. Generally, when the slope coefficient is biased downwards towards zero, the intercept, α, is made positive even if it is really zero.

2 *Simultaneous determination of inflation and exchange rates* It is another characteristic of the regression methodology that if the direction of causation goes from inflation to exchange rates and vice versa, then failure to use simultaneous-equation methods biases coefficients such as β, again usually towards zero.[31] In the case of PPP, causation does go both ways because changes in exchange rates affect inflation and inflation affects exchange rates. Neither side of the regression equation is predetermined, an essential characteristic required to avoid coefficient bias.

As mentioned, researchers who have tried to overcome the statistical problems, and who have considered inflation versus exchange rates over long time periods, have tended to support PPP. For example, by considering only the long-term trends which remain after removing "noise" in the data, Mark Rush and Steven Husted have shown that PPP holds for the US dollar versus other currencies.[32] The Rush and Husted results indirectly support the view that departures from PPP are due to poor measurement of inflation: long-term trends should reduce or remove the unsystematic errors in calculating inflation because the errors should average out and become relatively less important as the interval of measurement increases. Further support for the view that data errors are responsible for rejection of PPP is provided by Craig Hakkio who reduced the possible problem of poor inflation measurement by considering many countries concurrently over many periods.[33] This reduces the role of measurement errors by reducing the role of any one variable containing unsystematic errors. Hakkio is unable to reject PPP.

Further indication that errors in the inflation variable may be responsible for poor support for PPP is found in tests using **cointegration techniques**, which involve studying the differences between two variables versus the variables themselves. The basic intuition behind the cointegration method is that if two economic variables move together, then differences between them should be more stable than the original series. This means that if the spot exchange rate and the inflation differential do move together according to long-run PPP, then even though the two series may temporarily move apart, they must move back eventually through their cointegration. The cointegrating coefficient, which is the equivalent to β in equation (7.18), should equal unity if PPP holds.

Yoonbai Kim has applied cointegration methods to the PPP relationship of the US dollar versus the

30 See Maurice D. Levi, "Errors in the Variables Bias in the Presence of Correctly Measured Variables," *Econometrica*, September 1973, pp. 985–6, and Maurice D. Levi, "Measurement Errors and Bounded OLS Estimates," *Journal of Econometrics*, September 1977, pp. 165–71.

31 This bias exists if common factors affect $(\dot{P}_{US} - \dot{P}_{UK})/(1 + \dot{P}_{UK})$ and $\dot{S}(\$/£)$. See Maurice D. Levi, "World-Wide Effects and Import Elasticities," *Journal of International Economics*, May 1976, pp. 203–14.

32 See Mark Rush and Steven Husted, "Purchasing Power Parity in the Long Run," *Canadian Journal of Economics*, February 1985, pp. 137–45.

33 Craig S. Hakkio, "A Re-examination of Purchasing Power Parity: A Multi-Country and Multi-Period Study," *Journal of International Economics*, November 1984, pp. 265–77. See also Yoonbai Kim, "Purchasing Power Parity in the Long-Run: A Cointegration Approach," *Journal of Money, Credit and Banking*, November, 1990, pp. 491–503.

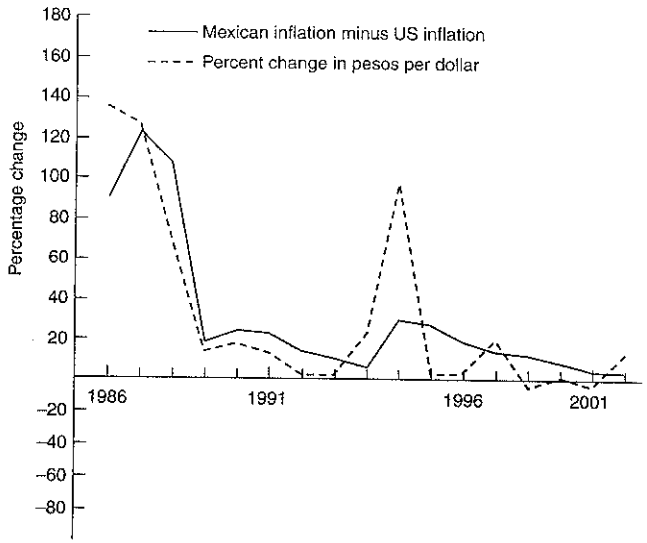


Figure 7.1 *US–Mexican inflation and the peso–dollar exchange rate*

Notes
The dotted line plots the percentage change in the spot rate of pesos per dollar, while the solid line shows Mexican inflation minus US inflation. The ups and downs of these series show that while the two series are different, they move closely enough together to provide casual support for PPP.

Source: International Financial Statistics, International Monetary Fund, Washington, DC.

currencies of Britain, France, Italy, Japan, and Canada.[34] He used annual data and found support for PPP using both consumer price indexes and wholesale price indexes in all cases except for Canada. Robert McNown and Myles Wallace have also applied cointegration techniques, focusing on the situation for countries with very high inflation rates.[35] This is a potentially fruitful context because the extreme values of changes in exchange rates and inflation differentials should minimize statistical problems such as the use of incorrectly constructed indexes. McNown and Wallace studied Israel, Argentina, Brazil, and Chile, all of which have had periods of very high inflation as well as periods of modest inflation. The estimates of cointegration coefficients were not significantly different from unity as hypothesized according to long-run PPP.

Casual support for PPP involving a country which has had periods of high inflation is provided by a simple plot of the spot rate between the Mexican peso and the US dollar versus the inflation differential between these countries. This is shown in Figure 7.1 for a period when Mexican inflation and the peso–dollar exchange rate varied widely. While the time paths of the two series do not track each other exactly from year to year, the overall correspondence is quite clear.

THE PRACTICAL IMPORTANCE OF PPP

It might seem that failure of PPP to hold would reduce its practical relevance. After all, it would then be of little use in helping to forecast exchange

34 See Yoonbai Kim, op. cit.

35 See Robert McNown and Myles S. Wallace, "National Price Levels, Purchasing Power Parity and Cointegration: A Test of Four High Inflation Economies," *Journal of International Money and Finance*, December 1989, pp. 533–46.

rates. In fact, the very opposite is true about PPP deviations. Knowing whether PPP holds, and knowing whether it holds more-or-less continuously or only on average over many years, is important in international managerial decisions. For example, if it is known that PPP is systematically violated and that these violations are unlikely to be corrected for a long or indefinite period of time, this knowledge can direct production decisions. Plants can be located where costs are lowest as well as on the basis of comparative advantage. This involves studying the costs in different locations to determine which currency is undervalued most relative to its PPP value, and where reversion is likely to be the slowest. On the other hand, if PPP holds more-or-less continuously, the only reason for choosing one production location over another is comparative advantage.

If PPP departures occur, but these are corrected over time, knowledge of how long such corrections might take is useful for determining whether it is worthwhile exploiting the temporary departures. Companies for which start-up and shut-down of activities is easiest will have the most to gain from paying attention to the behavior of PPP.

SUMMARY

1 The law of one price states that a commodity will have the same price in terms of a common currency in every country. The law follows from commodity arbitrage, which involves buying in the cheapest country if prices are different.

2 It follows from the law of one price that the dollar price of a commodity in the United States equals the pound price of the commodity in Britain multiplied by the spot exchange rate of dollars per British pound. Deviations from this relationship can be caused by transportation costs and import tariffs.

3 The principle of PPP is the extension of the law of one price to prices of a basket of goods. In its absolute form, PPP says that the dollar price of a basket of goods in the United States is the pound price of the basket in Britain, multiplied by the exchange rate of dollars per pound.

4 In its relative form, PPP says that the rate of change of the exchange rate is approximately equal to the difference between inflation rates.

5 Speculation and efficient markets also produce the relative form of the PPP condition in terms of expected values.

6 Empirical support for the PPP condition is weak, although there is some evidence it may hold in the long run.

7 The reasons the law of one price and PPP do not hold include transportation costs, tariffs, quotas, and the fact that there are goods and services that are nontradable. PPP may not hold even if the law of one price holds for every item because of different weights for different items in different countries' price indexes.

8 The fact that empirical evidence does not support PPP may be due to statistical difficulties in evaluating the principle.

REVIEW QUESTIONS

1 What is the "law of one price?"
2 How does the law of one price relate to PPP?
3 Explain what a commodity arbitrager does.
4 Write down the absolute form of PPP and interpret it.
5 Write down the relative form of PPP and interpret it.
6 Write down the efficient markets (or expectations) form of PPP and interpret it.
7 Why might the efficient markets (or expectations) form hold exactly even though the relative form of PPP does not?
8 What statistical problems are associated with empirical tests of PPP?

ASSIGNMENT PROBLEMS

1 Why might the law of one price hold even in the presence of import tariffs?
2 Assume that the prices of a standard basket of goods and services in different countries are as follows

United States	$400
Canada	C$550
United Kingdom	£280
Japan	¥60,000

   a What are the implied PPP exchange rates?
   b How would the presence of a high sales tax in Britain, such as the VAT, influence your guess of where the actual value of $S(\$/£)$ would be vis à vis the PPP value of $S(\$/£)$?
3 Why might there be departures from PPP even if the law of one price holds for every commodity?
4 a Assume $\dot{P}_{Mex} = 50$ percent and $\dot{P}_{US} = 2$ percent. Calculate $\dot{S}(P_S/\$)$ and $\dot{S}(\$/P_S)$ according to the precise and approximate dynamic PPP conditions.
   b Assume that Mexican inflation increases to 100 percent, while US inflation remains at 2 percent. Calculate $\dot{S}(P_S/\$)$ and $\dot{S}(\$/P_S)$ according to the precise and approximate PPP conditions.
   c How does the error in the approximate condition depend on whether you are measuring $\dot{S}(P_S/\$)$ or $\dot{S}(\$/P_S)$?
   d How would a constant percentage sales tax in Mexico affect your answers above?
5 If speculators are risk-averse, could this affect the accuracy of the link between the expected change in the exchange rate and the difference between expected inflation rates?
6 Is the accuracy of the approximate PPP condition negatively affected by the level of inflation?

7 Why might the relative form of PPP hold even though the absolute form does not hold?
8 Assume inflation in Brazil is 15 percent and in China 2 percent. What is the percent change in the exchange rate of Brazilian reals per Chinese RMB, and Chinese RMB per Brazilian real, and why do these two percent changes differ?
9 What is required for price discrimination between markets to cause departures from the law of one price?
10 What is the relevance of a "goodness of fit" measure such as the R^2 statistic, for judging PPP?
11 What is one-way versus two-way arbitrage in the context of PPP, and how do the implications of the two types of arbitrage differ?
12 Specialization, which has accompanied freer trade, has caused countries to produce larger amounts of each of a narrower range of products, trading these for the wider range of products that people consume. How might this have affected PPP?
13 What characteristics of a Big Mac® led *The Economist* to choose it as a basis for their alternative PPP exchange-rate measure? Can you suggest any other items which might be used?
14 Several possibilities, both theoretical and empirical, have been raised to explain the apparent failure of PPP. List and explain at least three of these. Comment on the validity of the explanation.
15 Does empirical evidence suggest PPP may be more likely to hold in the short-run or the long-run? Can you suggest an explanation for why this might be true?

BIBLIOGRAPHY

Abuaf, Niso and Philippe Jorion, "Purchasing Power Parity in the Long Run," *Journal of Finance*, March 1990, pp. 157–74.

Balassa, Bela, "The Purchasing-Power Parity Doctrine: A Re-Appraisal," *Journal of Political Economy*, December 1964, pp. 584–96. Reprinted in Richard N. Cooper (ed.), *International Finance: Selected Readings*, Penguin Books, Middlesex, UK, 1969.

Dornbusch, Rudiger and Dwight Jaffee, "Purchasing Power Parity and Exchange Rate Problems: Introduction" and the papers included in "Purchasing Power Parity: A Symposium," *Journal of International Economics*, May 1978, pp. 157–351.

Frenkel, Jacob A., "Exchange Rates, Prices and Money: Lessons from the 1920s," *American Economic Review*, March 1980, pp. 235–42.

Gaillot, Henry J., "Purchasing Power Parity as an Explanation of Long-Term Changes in Exchange Rates," *Journal of Money, Credit and Banking*, August 1970, pp. 348–57.

Hakkio, Craig, "A Re-examination of Purchasing Power Parity: A Multi-Country and Multi-period Study," *Journal of International Economics*, November 1984, pp. 265–77.

Huang, Roger, "Expectations of Exchange Rates and Differential Inflation Rates: Further Evidence on Purchasing Power Parity in Efficient Markets," *Journal of Finance*, March 1987, pp. 69–79.

—, "Risk and Parity in Purchasing Power," *Journal of Money, Credit and Banking*, August 1990, pp. 338–56.

Kim, Yoonbai, "Purchasing Power Parity in the Long Run: A Cointegration Approach," *Journal of Money, Credit and Banking*, November 1990, pp. 491–503.